AI Refinement of a Heath–Brown Type Zero-Free Region for the Riemann Zeta Function

From the 4.8913 Baseline to a 4.8038 Candidate

https://github.com/BlinkDL/VibeMath
I would say this is 99% AI, although I provided some inputs and guidance.

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Abstract

The original version of this manuscript described a certificate-based refinement of the Bellotti–Trudgian–Yang argument from the published constant 4.896 to the candidate constant 4.8913. This revised manuscript retains that calculation as a baseline and adds a supplement explaining the subsequent route to the internal candidate

$$\zeta(\sigma + it) \neq 0 \qquad \left(t \geq 3, \quad \sigma > 1 - \frac{1}{4.8038 \log t}\right).$$

The later refinement is not a logical consequence of the 4.8913 statement alone: it additionally imports the external classical seed 4.862 and the Littlewood constant 19.62, as recorded in work of Lee from results of Yang. The new internal computation has three main parts. First, a second explicit Fejér square $P_{12}^{(2)}$ and direct finite prime-power certificates raise the raw orbit floor from 0.2415849 to 0.2439742. Second, a non-circular finite bootstrap, with the auxiliary line recomputed from the currently known region at every step, gives the pre-Fourier candidate 4.8062105. Third, the signed shifts in the central explicit-formula error are combined before absolute values are taken. They collapse to one Fourier transform

$$K_r(v) = \widehat{g_r}(v), \qquad g_r(x) = e^{-rx/2} p(e^{-rx}) h(x),$$

and a Carlson–Sobolev estimate gives the uniform certificate

$$\|K_r\|_1 \leq \frac{2\pi \cdot 72}{r^2} \qquad (r \geq 136).$$

This nearly halves the dominant quadratic loss in the sharpened Lemma 8 ledger and yields the finite chain

$$4.806212 \longrightarrow 4.8043 \longrightarrow 4.8040 \longrightarrow 4.8039 \longrightarrow 4.8038.$$

The result remains computer-assisted and should be regarded as an internally closed candidate rather than a published theorem. The exact rational, Sturm, Bernstein, and moment computations are separated from the directed interval computations, and an independent Arb or MPFI replay is still required.

1 Introduction

For $s = \sigma + it$ with $\sigma > 1$, the logarithmic derivative

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n \geq 1} \frac{\Lambda(n)}{n^s}$$

turns nonnegative prime-power coefficients into a zero detector. Classical arguments combine this positivity with a nonnegative trigonometric polynomial. Stechkin's use of the functional equation and Heath–Brown's compactly supported smoothing function make this detector substantially stronger. Bellotti, Trudgian and Yang [1] combine these two ideas and obtain

$$\sigma > 1 - \frac{1}{4.896 \log t}, \qquad t \geq 3.$$

Their proof has three numerically sensitive components:

(i) the nonnegative trigonometric polynomial;

(ii) a lower bound for grouped prime-power orbits;

(iii) the finite fixed-point ledger after the explicit formula is applied.

The present note modifies all three components, but keeps the external analytic inputs of [1] unchanged: Ford's estimate for the Laplace transform, the smoothed explicit formula, the zero-counting estimates, the verification of the Riemann hypothesis up to $3 \cdot 10^{12}$, and Yang's explicit Littlewood region. The genuinely new computations are:

(1) an exact degree-twelve Fejér square P_{12};

(2) exact prime-power lower bounds for every prime $p < 100$;

(3) a direct interval treatment of the dominant $m = 0$ Laplace integral;

(4) a corrected and sharpened remainder ledger;

(5) the final fixed-point calculation for $R = 4.8913$.

We formulate the main statement in certificate-dependent form.

Theorem 1.1 (Computer-assisted refinement)**.** *Assume the external explicit estimates used in [1], including the verification of the Riemann hypothesis up to $H = 3 \cdot 10^{12}$ and the explicit Littlewood region*

$$\sigma > 1 - \frac{\log \log t}{21.233 \log t}.$$

If the exact rational certificates and directed interval enclosures described in Sections 4–6 are replayed, then

$$\zeta(\sigma + it) \neq 0 \qquad \left(t \geq 3, \quad \sigma > 1 - \frac{1}{4.8913 \log t} \right).$$

Verification status 1.2. The arithmetic certificates use exact integers and rational numbers. The remaining one-dimensional integrals and parameter inequalities use directed interval arithmetic. The note records the full mathematical reduction and the numerical margins, but it is not a proof-assistant formalization. An independent Arb, MPFI, or comparable replay is therefore desirable.

2 The smoothing function and normalization

We retain the smoothing function of [1]. Fix

$$\theta = 1.1338, \qquad U = 2\theta \cot \theta,$$

and define, for $0 \le u \le U$,

$$w(u) = \sec^2\theta \Bigg(\sec^2\theta \left(\theta\cot\theta - \frac{u}{2}\right)\cos(u\tan\theta) + 2\theta\cot\theta - u + \frac{\sin(2\theta - u\tan\theta)}{\sin 2\theta} - 2\left(1 + \frac{\sin(\theta - u\tan\theta)}{\sin\theta}\right)\Bigg), \tag{1}$$

with $w(u) = 0$ outside $[0, U]$. Numerically,

$$U = 1.0592329388789857\ldots, \qquad w(0) = 5.6727875988616045\ldots.$$

Let

$$(\kappa_0, \ldots, \kappa_6) = \left(1, -\frac{851}{859}, \frac{780}{859}, -\frac{525}{859}, \frac{171}{859}, \frac{28}{859}, -\frac{29}{859}\right),$$

so that

$$\kappa := \sum_{m=0}^{6} \kappa_m = \frac{433}{859}.$$

For $\delta_m = (2\sigma - 1)m$, define

$$f(u) = \eta w(\eta u) \sum_{m=0}^{6} \kappa_m e^{-\delta_m u}, \qquad F(z) = \int_0^\infty e^{-zu} f(u)\, \mathrm{d}u.$$

To avoid a harmless normalization ambiguity in displayed formulas, we set

$$f_*(0) := \eta w(0).$$

Thus the literal value of the function in (2) is $f(0) = \kappa f_*(0)$, while the explicit-formula ledger is most naturally written in units of $f_*(0)$.

Two elementary positivity facts are useful. First, if

$$p(x) = \sum_{m=0}^{6} \kappa_m x^m,$$

then

$$859p(x) = \sum_{j=0}^{6} b_j \binom{6}{j} x^j (1 - x)^{6-j}, \tag{2}$$

where

$$(b_0, \ldots, b_6) = \left(859, \frac{4303}{6}, \frac{1882}{3}, \frac{2253}{4}, \frac{7651}{15}, 469, 433\right).$$

Hence

$$p(x) \ge \frac{433}{859} = \kappa \qquad (0 \le x \le 1).$$

Second, w is decreasing. Indeed, after putting $v = u\tan\theta$, direct simplification gives

$$w'(u) = \sec^2\theta \,\sin v \left[\tan\theta + \sec^2\theta\left(\frac{v}{2} - \theta\right) - \tan\frac{v}{2}\right].$$

Convexity of tan on $[0, \theta]$ makes the bracket nonpositive, so $w'(u) \le 0$.

For the final parameter range, directed interval subdivision verifies

$$f(u) \ge \kappa\eta w(0) \qquad (0 \le u \le 59). \tag{3}$$

3 The degree-twelve Fejér square

Define

$$\begin{aligned}Q_{12}(z) = 32635 &- 54033z - 17946z^2 + 174507z^3 + 66931z^4 - 374536z^5 \\ &- 168219z^6 + 1652944z^7 + 4446322z^8 + 6169255z^9 + 5432436z^{10} \\ &+ 3003044z^{11} + 837164z^{12}. \end{aligned} \tag{4}$$

Its squared Euclidean norm is

$$\mathcal{N} := \sum_{j=0}^{12} q_j^2 = 100000002139070.$$

We take

$$P_{12}(x) = \frac{|Q_{12}(e^{ix})|^2}{\mathcal{N}} = 1 + \sum_{k=1}^{12} a_k \cos(kx). \tag{5}$$

Nonnegativity is automatic. The exact numerators $\mathcal{N}a_k$ are

$$\begin{aligned}(&173779234139896, 111942599517052, 49767351006488, 12146409523234,\\ &9367972, 9065744, 1076858103392, 588712579182, 2346122, 654528,\\ &105539717056, 54641694280).\end{aligned}$$

Every entry is positive. The principal parameters are

$$a_1 = 1.737792304226366158\ldots, \tag{6}$$

$$a := \sum_{k=1}^{12} a_k = 3.494613602397228815\ldots, \tag{7}$$

$$P_{12}(0) = 1 + a = 4.494613602397228815\ldots. \tag{8}$$

The polynomial was obtained by a constrained numerical search, followed by integer rationalization. Only the displayed integer vector is used in the proof.

4 Exact prime-power orbit certificates

For a prime p and an integer $M \geq 1$, set

$$G_{p,M}(\sigma, x) = \sum_{m=1}^{M} p^{-m\sigma} P_{12}(mx).$$

Since $P_{12} \geq 0$,

$$\frac{\partial}{\partial\sigma} G_{p,M}(\sigma, x) = -(\log p) \sum_{m=1}^{M} m p^{-m\sigma} P_{12}(mx) \leq 0. \tag{9}$$

Thus $G_{p,M}(\sigma, x) \geq G_{p,M}(1, x)$ whenever $\sigma \leq 1$.

For strict compatibility with the support range in (3), use

$$M_p = \min\left(15, \left\lfloor \frac{59}{\log p} \right\rfloor\right).$$

The infinite orbit is explicit. Writing $y = \cos x$ and denoting Chebyshev polynomials by T_k, one has

$$G_{p,\infty}(1, x) = \frac{1}{p-1} + \sum_{k=1}^{12} a_k \frac{pT_k(y) - 1}{p^2 - 2pT_k(y) + 1}. \tag{10}$$

Moreover,

$$0 \le G_{p,\infty}(1,x) - G_{p,M_p}(1,x) \le \frac{P_{12}(0)}{(p-1)p^{M_p}}. \tag{11}$$

For each prime $p < 100$, subtracting a rational candidate b_p, applying (11), and multiplying by the twelve positive denominators in (10) gives an integer polynomial $R_p(y)$ of degree at most 78. On replacing y by $2u-1$, R_p is certified positive on $[0,1]$ by exact dyadic Bernstein subdivision. The resulting lower bounds are shown in Table 1.

p	b_p	p	b_p
2	0.2383955346	43	0.0000071512
3	0.0518827925	47	0.0000054858
5	0.0075903925	53	0.0000038329
7	0.0021992359	59	0.0000027820
11	0.0004621014	61	0.0000025181
13	0.0002683487	67	0.0000019018
17	0.0001156281	71	0.0000015988
19	0.0000822567	73	0.0000014712
23	0.0000461824	79	0.0000011612
29	0.0000230998	83	0.0000010015
31	0.0000189391	89	0.0000008126
37	0.0000111868	97	0.0000006279
41	0.0000082412		

Table 1: Certified lower bounds for $G_{p,M_p}(1,x)$.

The logarithms are also bounded from below by rational arithmetic, using the positive truncation

$$\log r = 2\sum_{j=0}^{J-1} \frac{u^{2j+1}}{2j+1} + \text{positive remainder}, \qquad u = \frac{r-1}{r+1},$$

after extracting a power of two. This yields

$$\sum_{p<100} (\log p) b_p > 0.2415849. \tag{12}$$

The high-precision value is

$$0.241584915780693204849895809 4\ldots.$$

Proposition 4.1 (Improved arithmetic term)**.** *For the smoothing function in Section 2 and the polynomial P_{12},*

$$\sum_{n\ge1} \frac{\Lambda(n)}{n^\sigma} f(\log n) P_{12}(t\log n) > 0.1217767\, \eta w(0). \tag{13}$$

Proof. Group the terms with $n = p^m$, use (3), (9), and the orbit certificates:

$$\begin{aligned}\sum_{n\ge1} \frac{\Lambda(n)}{n^\sigma} f(\log n) P_{12}(t\log n) &\ge \kappa\eta w(0) \sum_{p<100} (\log p) G_{p,M_p}(1, t\log p) \\ &> \frac{433}{859} \cdot 0.2415849\, \eta w(0) \\ &> 0.1217767\, \eta w(0).\end{aligned}$$

□

The corresponding constant in [1] is 0.1186. Thus the arithmetic part alone improves by approximately 2.68%.

5 Recomputed analytic ledger

We now record only the changes needed in the explicit-formula calculation. The notation follows [1]. Put

$$\mu = \frac{1-\sigma}{\eta}, \qquad T_j(\sigma) = \sum_{m=0}^{6} \frac{\kappa_m}{(1+(2\sigma-1)m)^j}.$$

Clearing denominators shows algebraically that

$$T_1'(\sigma) > 0, \qquad T_2'(\sigma) > 0 \qquad (\sigma \geq 1/2).$$

No floating-point root isolation is required.

The reciprocal expansion is

$$W\left(\frac{\delta_m}{\eta} + x\right) = \frac{w(0)}{\delta_m}\eta - \frac{xw(0)}{\delta_m^2}\eta^2 + E_m(\eta). \tag{14}$$

For $x = -\mu$ and $x = 1-\mu$, the signs in the quadratic terms must be read directly from (14). Uniform estimates use $|x| \leq 1$, and Ford's parameter is scaled by η^{-1}. With these choices, the combined $m \geq 1$ terms and the $F(\sigma-\eta)$ term satisfy the following safe bound throughout the final parameter box:

$$\begin{aligned} a_1 F(\sigma - 1 + \eta) + a_1 F(\sigma - \eta) - F(\sigma - 1) \\ \geq I_Q(\mu) + 3.9090\,\eta + 20.189\,\eta^2 - 170\,\eta^3, \end{aligned} \tag{15}$$

where

$$I_Q(\mu) = \int_0^U e^{\mu u}\big(a_1 e^{-u} - 1\big) w(u)\, du. \tag{16}$$

The direct treatment of I_Q is important. The cubic Taylor minorant used for a coarse uniform estimate loses more than the final fixed-point margin. On the actual interval needed in Section 6, directed interval integration gives

$$I_Q(\mu) > 0.99348, \qquad 0.136 < I_Q'(\mu) < 0.138. \tag{17}$$

The gamma-factor estimate gives a gain greater than $1.5630\,\eta w(0)$. Combining this with Proposition 4.1, the explicit-formula errors

$$53a\,\eta^2 + (2353 + 1601a)\eta^3,$$

and (15), one obtains

$$\frac{a\kappa w(0)}{2}\,\eta \log t \geq I_Q(\mu) + C_Q(\eta) - 10^{-7}, \tag{18}$$

where the safe cubic polynomial

$$C_Q(\eta) = 13.4663\,\eta - 165.026\,\eta^2 - 8117.877\,\eta^3 \tag{19}$$

may be used. Finally,

$$\frac{a\kappa w(0)}{2} = 4.9964370814383408\ldots < 4.99644. \tag{20}$$

For completeness, a separate interval audit of the remaining numerical lemmas gives the following safe statements in the present parameter range:

- the $t = 0$ integral occurring in the gamma error is less than 34.652; the two contributions still sum to less than $309\eta^3 < 623\eta^3$;

- for $t \geq H$, the error remains bounded by $14\eta^2 + 424\eta^3$;
- the far-zero loss remains below 10^{-8} per detector and below 10^{-7} after the trigonometric weights are summed;
- the terms $F(s_k - 1)$ with $k \geq 1$ are negligible compared with the published $10^{-10}\eta$ allowance.

6 The fixed-point calculation

Set

$$R = 4.8913, \qquad A_0 = \frac{1}{R} = 0.2044446261730010\ldots,$$
$$H = 3 \cdot 10^{12}, \qquad T_0 = 10^{10}, \qquad K = 12.$$

Define

$$\eta_0 = \frac{A_0}{\log H} = 0.007116158542429939\ldots,$$
$$\sigma_0 = 1 - \frac{A_0}{\log(KH + T_0)} = 0.993450398220328925\ldots.$$

In particular,

$$\frac{2\sigma_0 - 1}{\eta_0} = 138.6844869400089\ldots > 138,$$

so the auxiliary Ford and nonnegativity estimates remain inside their certified ranges.

The explicit Littlewood region takes over once

$$\log t \geq L_* := \exp(21.233 A_0) = 76.78219271137694\ldots.$$

It is therefore enough to treat

$$H \leq t \leq e^{L_*}.$$

Under the usual contradiction hypothesis for the iteration lemma,

$$\eta > \frac{1}{6\log t}.$$

Moreover, throughout this interval one may use

$$\mu \geq 1 - \frac{c_K}{\log t}, \qquad c_K = \log\left(12 + \frac{1}{300}\right) + 10^{-6} = 2.485185388992674\ldots.$$

At the worst endpoint $\log t = L_*$ this gives

$$\mu_* = 0.9676333105211719\ldots, \tag{21}$$
$$\eta_* = \frac{1}{6L_*} = 0.002170642186439818\ldots. \tag{22}$$

The function

$$\mathcal{B}(L) = I_Q\left(1 - \frac{c_K}{L}\right) + C_Q\left(\frac{1}{6L}\right)$$

is decreasing on $[\log H, L_*]$. Indeed,

$$L^2\mathcal{B}'(L) = c_K I_Q'(\mu) - \frac{1}{6}C_Q'(\eta),$$

and the interval bounds in (17), together with the explicit polynomial (19), make the right-hand side strictly negative. Thus the endpoint is the only value that must be checked.

At that endpoint,

$$I_Q(\mu_*) > 0.99348, \qquad C_Q(\eta_*) > 0.02836.$$

Using (18) and (20),

$$\begin{aligned}\eta \log t &> \frac{0.99348 + 0.02836 - 10^{-7}}{4.99644} \\ &> 0.2045135 \\ &> 0.2044446261730010 = A_0.\end{aligned}$$

This contradicts the assumption that a zero satisfies

$$(1-\beta)\log t < A_0 + 10^{-100}.$$

Hence the iteration advances by 10^{-100} until the value A_0 is reached. The verified-zero computation below H and Yang's Littlewood region above e^{L_*} complete the proof of Theorem 1.1, subject to the certificate status stated after the theorem.

7 Supplement: from 4.8913 to 4.8038

The preceding sections reproduce the original 4.8913 baseline. The present supplement records the later refinements. It is important to distinguish logical dependence from methodological continuity.

Remark 7.1 (Additional external inputs). The 4.8038 candidate is not obtained from the 4.8913 region alone. The later bootstrap also uses the external classical seed

$$\sigma > 1 - \frac{1}{4.862 \log t}$$

and the Littlewood region with constant 19.62. These constants are recorded by Lee [7] as consequences of Yang's work. Thus the supplement refines the computational method of the 4.8913 manuscript, but it imports a stronger starting region.

Theorem 7.2 (Supplementary certificate statement)**.** *Assume the external analytic and verified-zero inputs used in [1], together with the classical seed* 4.862 *and the Littlewood constant* 19.62 *recorded in [7]. Assume also that the exact rational, Sturm, Bernstein, finite-moment, and directed-interval certificates described in this supplement are replayed. Then the supplemented computation gives*

$$\zeta(\sigma + it) \neq 0 \qquad \left(t \geq 3, \quad \sigma > 1 - \frac{1}{4.8038 \log t}\right).$$

Verification status 7.3. The statement is an internally closed computer-assisted candidate. The finite algebraic certificates are exact, while the one-dimensional integral and fixed-point bounds use directed interval arithmetic. Neither an independent Arb/MPFI replay nor a third-party line-by-line review has yet been completed. The Yang thesis itself was not independently audited here; the constants 4.862 and 19.62 are used through the explicit report in [7].

7.1 The correct non-circular finite bootstrap

Let $R_{\rm s}$ be a zero-free constant already proved and let $R_{\rm t} < R_{\rm s}$ be the stronger target. It is convenient to work with widths

$$A = \frac{1}{R_{\rm s}}, \qquad B = \frac{1}{R_{\rm t}}, \qquad B > A.$$

If a hypothetical zero is written as

$$\rho_0 = \beta_0 + it, \qquad \eta = 1 - \beta_0, \qquad L = \log t,$$

then the seed region and the target contradiction hypothesis give

$$\eta L \geq A, \qquad \eta L < B. \tag{23}$$

The auxiliary line must be defined from the already known width A:

$$\sigma = 1 - \frac{A}{\log(Ke^L + T_0)}. \tag{24}$$

It is not legitimate to substitute the target width B into (24) before B has been established. Put

$$c_H = \log\left(K + \frac{T_0}{H}\right).$$

From (23) and $\log(Ke^L + T_0) \leq L + c_H$ one obtains

$$\eta \geq \frac{A}{L}, \qquad \mu := \frac{1-\sigma}{\eta} > \frac{A}{B}\frac{L}{L + c_H}. \tag{25}$$

The factor A/B in the second inequality is essential.

Let $D_Q = a\kappa w(0)/2$ be the normalizing denominator and let $C_{1,Q}$ and $C_{2,Q}$ denote the certified lower-bound functions after the explicit formula is assembled. A finite step $A \to B$ is proved once

$$\mathcal{L}_{A,B}(L) := \frac{C_{1,Q}\left(\frac{A}{B}\frac{L}{L+c_H}\right) + C_{2,Q}(A/L) - 10^{-7}}{D_Q} > B \tag{26}$$

throughout

$$\log H \leq L \leq \exp(19.62B).$$

Above the right endpoint the Littlewood region is already stronger than the target classical region. In every step used below, interval differentiation shows that the left-hand side of (26) is decreasing, so only the right endpoint needs to be checked.

This finite-step formulation replaces a formally circular fixed-point substitution by a genuine induction in which every target uses only a previously certified seed.

7.2 A second Fejér square and direct finite prime-power certificates

The first large improvement after the 4.8913 baseline comes from replacing (4) by

$$\begin{aligned} Q_{12}^{(2)}(z) = 2973020 &- 5101549z - 1556727z^2 + 17311464z^3 + 5669545z^4 \\ &- 38313778z^5 - 13979425z^6 + 174340757z^7 + 455260827z^8 \\ &+ 620102516z^9 + 535373941z^{10} + 288437326z^{11} + 77213430z^{12}. \end{aligned} \tag{27}$$

Its squared norm is

$$\mathcal{N}_2 = 1000000000489220451,$$

and

$$P_{12}^{(2)}(x)=\frac{|Q_{12}^{(2)}(e^{ix})|^2}{\mathcal{N}_2}=1+\sum_{k=1}^{12}a_k^{(2)}\cos(kx). \tag{28}$$

The exact numerators $\mathcal{N}_2 a_k^{(2)}$ are

$$\begin{aligned}(&1736607730982643766, 1116308926917508272, 494498266956404318,\\ &120058469375169958, 428646472, 314671138, 10351059402375478,\\ &5575245011467526, 258706458, 101620472, 927243684682900, 459114143317200).\end{aligned}$$

Every entry is positive. In particular,

$$a_1^{(2)}=1.736607730133059749\ldots,\qquad \sum_{k=1}^{12}a_k^{(2)}=3.484786055872385352\ldots.$$

For a prime p and

$$M_p=\min\left(15,\left\lfloor\frac{57.5}{\log p}\right\rfloor\right),$$

consider the finite orbit

$$G_{p,M_p}^{(2)}(x)=\sum_{m=1}^{M_p}p^{-m}P_{12}^{(2)}(mx).$$

Unlike the baseline calculation, no infinite orbit and no crude geometric tail are introduced. After setting $y=\cos x$ and $y=2u-1$, clearing the positive factor $\mathcal{N}_2 p^{M_p}$ turns $G_{p,M_p}^{(2)}(x)-b_p$ into an integer polynomial in u of degree at most $12M_p\le 180$. Exact dyadic de Casteljau subdivision proves positivity of its Bernstein coefficients on every terminal subinterval. The result is

$$\boxed{\sum_{p<100}(\log p)b_p>0.2439742.} \tag{29}$$

The previous raw floor was 0.2415849, so the gain at this stage is about 0.989%.

For the last pre-Fourier bootstrap vector

$$\begin{aligned}10^{10}(\kappa_0,\ldots,\kappa_6)=\big(10^{10},-9986207037,9793152105,-8692402447,\\ 6137992779,-2719409222,501106985\big).\end{aligned} \tag{30}$$

we have

$$\kappa=0.5034233163,$$

and hence the certified arithmetic contribution is

$$\boxed{\kappa\cdot 0.2439742=0.12282230085563946\ldots.} \tag{31}$$

The corresponding baseline value in Proposition 4.1 was 0.1217767.

7.3 The pre-Fourier stage: reaching 4.8062105

The coefficients κ_m are reoptimized subject to four families of exact constraints:

(i) Bernstein positivity and monotonicity of $p(x)=\sum\kappa_m x^m$ on $[0,1]$;

(ii) global positivity of the Lemma 5 rational function after clearing denominators, certified by a Sturm sequence on $X\ge 0$;

(iii) coefficientwise negativity of the cleared numerators proving $T_1'(\sigma),T_2'(\sigma)>0$;

(iv) directed-interval versions of the smoothing, Ford-remainder, gamma, and fixed-point inequalities.

At the numerical optimum the scaled Lemma 5 function $(1+X)B_\kappa(X)$ has three nearly equal local minima,

$$X_1 \approx 0.635099, \qquad X_2 \approx 15.18344, \qquad X_3 \approx 357.68873.$$

The six approximate equations consisting of equal values and vanishing derivatives at these three points explain why the degree-six vector is almost rigid.

Using the seed 4.862, the Littlewood constant 19.62, the polynomial $P_{12}^{(2)}$, and stepwise rationalized κ-vectors gives the following non-circular chain. Thus the new arithmetic polynomial,

seed R_{s}	target R_{t}	directed-interval margin
4.862	4.8136	$6.17875422 \times 10^{-6}$
4.8136	4.80719	$1.90908374 \times 10^{-7}$
4.80719	4.8064	$3.04840982 \times 10^{-6}$
4.8064	4.80625	$8.64922615 \times 10^{-7}$
4.80625	4.80622	$2.62874828 \times 10^{-7}$
4.80622	4.806218	$3.70844830 \times 10^{-7}$
4.806218	4.806212	$5.20349606 \times 10^{-8}$
4.806212	4.8062105	$1.02037588 \times 10^{-8}$

Table 2: The finite bootstrap before the signed-shift improvement.

the exact finite orbits, the actual Ford constants, and the corrected finite-step logic together move the internal candidate from the 4.8913 baseline to 4.8062105.

7.4 Signed-shift Fourier collapse

The last improvement comes from postponing absolute values in the central explicit-formula remainder. Put

$$d = 2\sigma - 1, \qquad r = \frac{d}{\eta}, \qquad D = 2\theta\cot\theta.$$

In the notation of the BTY Laplace-transform estimate, write

$$W_0(z) = W(z) - \frac{w(0)}{z}.$$

Define

$$h(x) = \begin{cases} w(x) - w(0), & 0 \le x \le D, \\ -w(0), & x > D. \end{cases} \tag{32}$$

Since

$$\frac{w(0)}{z} = \int_0^\infty e^{-zx} w(0)\,\mathrm{d}x,$$

we have the exact representation

$$W_0(z) = \int_0^\infty e^{-zx} h(x)\,\mathrm{d}x. \tag{33}$$

The central error contains the signed combination

$$K_r(v) = \sum_{m=0}^{6} \kappa_m W_0\left(r\left(m + \frac{1}{2}\right) + iv\right). \tag{34}$$

Let

$$p(u) = \sum_{m=0}^{6} \kappa_m u^m.$$

Substituting (33) into (34) and retaining the signs until after summation gives

$$K_r(v) = \int_0^\infty e^{-ivx} e^{-rx/2} \sum_{m=0}^{6} \kappa_m e^{-rmx} h(x)\,\mathrm{d}x \tag{35}$$

$$= \int_0^\infty e^{-ivx} g_r(x)\,\mathrm{d}x, \tag{36}$$

where

$$\boxed{g_r(x) = e^{-rx/2} p(e^{-rx}) h(x).} \tag{37}$$

Thus seven shifted transforms collapse to one Fourier transform $K_r = \widehat{g}_r$.

Differentiating (37) introduces

$$q(u) = \frac{1}{2}p(u) + up'(u) = \sum_{m=0}^{6} \left(m + \frac{1}{2}\right) \kappa_m u^m, \tag{38}$$

and

$$g_r'(x) = e^{-rx/2} \left[p(e^{-rx})h'(x) - rq(e^{-rx})h(x)\right]. \tag{39}$$

For the seed vector (30), exact Bernstein coefficients prove

$$p(u) > 0, \qquad q(u) > 0 \qquad (0 \le u \le 1). \tag{40}$$

The same check is repeated for every rationalized vector in the final bootstrap.

7.5 Finite rational moments and the Carlson–Sobolev bound

Let

$$c = -w''(0).$$

Directed interval arithmetic gives

$$c < 51.74, \qquad \sup_{0 \le x \le D} |w'''(x)| < 144. \tag{41}$$

Since $h(0) = h'(0) = 0$ and $h''(0) = -c$, the certified Taylor bounds are

$$\left|h(x) + \frac{c}{2}x^2\right| \le 24x^3, \qquad |h'(x) + cx| \le 72x^2. \tag{42}$$

The extension beyond D is checked at the endpoint and then follows from the polynomial growth of the right-hand sides.

Put $y = rx$. Four quantities control the two Sobolev norms:

$$I_4 = \int_0^\infty e^{-y} p(e^{-y})^2 y^4\,\mathrm{d}y, \tag{43}$$

$$I_6 = \int_0^\infty e^{-y} p(e^{-y})^2 y^6\,\mathrm{d}y, \tag{44}$$

$$J = \int_0^\infty e^{-y} \left(\frac{1}{2}q(e^{-y})y^2 - p(e^{-y})y\right)^2 \mathrm{d}y, \tag{45}$$

$$E = \int_0^\infty e^{-y} \left(\frac{1}{2}p(e^{-y})y^2 + \frac{1}{6}q(e^{-y})y^3\right)^2 \mathrm{d}y. \tag{46}$$

They are finite rational sums, not numerical quadratures. For example,

$$I_j = j! \sum_{m,n=0}^{6} \frac{\kappa_m \kappa_n}{(m+n+1)^{j+1}}. \tag{47}$$

For the seed vector,

$$I_4 < 22.73, \qquad I_6 < 709.61, \tag{48}$$

$$J < 0.474, \qquad E < 19.96. \tag{49}$$

Consequently,

$$\|g_r\|_2 \le r^{-5/2} \left(\frac{c}{2}\sqrt{I_4} + \frac{24}{r}\sqrt{I_6} \right), \tag{50}$$

$$\|g_r'\|_2 \le r^{-3/2} \left(c\sqrt{J} + \frac{144}{r}\sqrt{E} \right). \tag{51}$$

Replaying the exact moment sums for the stepwise vectors gives, uniformly for $r \ge 136$,

$$\sqrt{A_r B_r} < 71.879405, \tag{52}$$

where A_r, B_r denote the bracketed quantities in (50)–(51).

For the Fourier convention $\widehat{g}(v) = \int_{\mathbb{R}} e^{-ivx} g(x)\,\mathrm{d}x$, Cauchy–Schwarz and Plancherel give, for any $a > 0$,

$$\|\widehat{g}\|_1^2 \le 2\pi^2 \left(a\|g\|_2^2 + \frac{\|g'\|_2^2}{a} \right).$$

Choosing $a = \|g'\|_2/\|g\|_2$ yields the Carlson–Sobolev inequality

$$\|\widehat{g}\|_1 \le 2\pi\sqrt{\|g\|_2\|g'\|_2}. \tag{53}$$

Combining (52) and (53) gives the safe uniform certificate

$$\boxed{\|K_r\|_1 \le \frac{2\pi \cdot 72}{r^2} \qquad (r \ge 136).} \tag{54}$$

7.6 The improved central explicit-formula error

The symmetric central interval in the BTY error term satisfies

$$\left| \sum_{m=0}^{6} \kappa_m E_{2,\mathrm{central}}(s + \delta_m) \right| \le \frac{\eta \log t}{2\pi} \|K_r\|_1. \tag{55}$$

If the current target window gives $\eta \log t \le A_{\mathrm{tar}}$, then $r = d/\eta$ and (54) imply

$$\boxed{\left| \sum_{m} \kappa_m E_{2,\mathrm{central}}(s + \delta_m) \right| \le \frac{72 A_{\mathrm{tar}}}{d^2} \eta^2.} \tag{56}$$

At the final parameters the coefficient in (56) is approximately

$$15.3961.$$

The previous sharpened ledger, after taking seven absolute values separately, used approximately

$$29.4452.$$

Thus the central quadratic loss is almost halved. Boundary and cubic terms are not reoptimized here; they retain the previously certified conservative bounds.

seed R_{s}	target R_{t}	directed-interval margin
4.806212	4.8043	1.18546×10^{-5}
4.8043	4.8040	1.16384×10^{-5}
4.8040	4.8039	8.18572×10^{-6}
4.8039	4.8038	3.29239×10^{-6}

Table 3: The signed-shift finite bootstrap.

7.7 The final bootstrap to 4.8038

The pre-Fourier stage reaches 4.8062105. For the signed-shift stage we use the slightly weaker certified seed 4.806212 and the following finite chain. The first row uses (30). The later rows use the rational vectors

$$\begin{aligned}
10^{10}\kappa^{(2)} &= \big(10^{10}, -9991606743, 9851742295, -8933214378, \\
&\qquad 6558980863, -3049313048, 597531062\big), && (57) \\
10^{10}\kappa^{(3)} &= \big(10^{10}, -9991606724, 9851742454, -8933211200, \\
&\qquad 6558980641, -3049309663, 597526570\big), && (58) \\
10^{10}\kappa^{(4)} &= \big(10^{10}, -9991606879, 9851744444, -8933217397, \\
&\qquad 6558995470, -3049319800, 597527505\big). && (59)
\end{aligned}$$

Each vector is rechecked independently for Bernstein positivity of p and q, Lemma 5 Sturm positivity, $T_1', T_2' > 0$, Ford remainders, smoothing positivity, and the finite-step criterion (26). This proves the supplementary certificate statement, subject to its stated status.

A target 4.8037 was also tested. The best proof slack found in the current search was about -8×10^{-6}. This is not a no-go theorem, but it shows that the present degree-six κ family, $Q_{12}^{(2)}$, smoothing parameter, and $C = 72$ Fourier certificate are close to a local wall.

8 Comparison of the two stages

The numerical change from 4.8913 to 4.8038 is

$$\frac{4.8913 - 4.8038}{4.8913} \cdot 100 = 1.7889\% \quad \text{in } R,$$

and

$$\left(\frac{4.8913}{4.8038} - 1\right) \cdot 100 = 1.8215\% \quad \text{in the width } 1/R.$$

The gain does not come from one parameter. It decomposes as follows.

(i) The baseline 4.8913 manuscript supplies the certificate architecture, the first Q_{12}, the initial prime-power grouping, and the corrected Laplace-transform ledger.

(ii) The stronger external constants 4.862 and 19.62 permit a wider, but still non-circular, finite bootstrap.

(iii) The second Fejér square and direct finite orbit certification increase the raw arithmetic floor to 0.2439742.

(iv) Reoptimized degree-six κ vectors and actual Ford constants give the pre-Fourier candidate 4.8062105.

(v) The signed-shift Fourier collapse eliminates a substantial loss caused by taking seven absolute values separately and gives 4.8038.

The central methodological principle is the same one that appears in Selberg's positive-source argument: combine the algebraically related terms before applying the triangle inequality. In the present setting the combination is not a Dirichlet convolution but the Fourier identity (36).

9 Updated reproducibility protocol

The supplemented computation separates into four layers.

Exact integer and rational arithmetic

1. Verify the norm and autocorrelations of (27).
2. For every $p < 100$, form the finite degree-$12M_p$ orbit polynomial and certify the chosen b_p by exact Bernstein subdivision.
3. For every bootstrap vector κ, certify Bernstein positivity of p and q, Sturm positivity of the Lemma 5 numerator, and coefficientwise signs for the derivatives of T_1 and T_2.
4. Evaluate the moments (43)–(46) as exact finite rational sums.

Directed interval arithmetic

1. Enclose $w''(0)$ and $\sup |w'''|$, then prove (41)–(42).
2. Replay the $C = 72$ Carlson bound for every step vector.
3. Check the smoothing-function lower bounds, Ford remainders, gamma terms, and finite-step monotonicity.
4. Verify every margin in Tables 2 and 3.

External analytic inputs

The replay imports the smoothed explicit formula, zero-counting estimates, Ford transform bounds, the verified-zero height, and the two Yang zero-free constants as reported in [1, 7]. These are not reproved here.

Still missing

A publication-quality result requires an independent interval engine, a direct audit of the precise hypotheses behind the 4.862 and 19.62 inputs, a continuous rewrite of the modified BTY lemmas, and third-party review.

10 Concluding assessment

The supplement produces a materially stronger internal candidate than the 4.8913 baseline, but it remains an exercise in the optimization of a classical zero-free-region argument. It does not produce a fixed half-plane, does not alter the asymptotic shape $1 - c/\log t$, and does not constitute a scale-level advance toward the Riemann hypothesis.

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